

BUAN 301

Session 5 : Conditional Probability, Independence, Bayes' Theorem

(9th, 11th Sep, 2026)

Agenda

❑ Joint and Marginal Probability

❑ Conditional Probability

❑ Independence of Events

❑ Bayes' Theorem

Joint & Marginal Probabilities

- Joint Probability is the probability of two events happening together
 - Given by $P(A \cap B)$

- Marginal Probability is given by the sum of the relevant Joint Probabilities

Illustration

□ Classroom example

Conditional Probability

□ Conditional Probability is the probability of an event given another event has occurred

□ Denoted by $P(A|B)$ – probability of event A given that B has occurred

Conditional Probability

□ Given two events A and B the conditional probability is the probability of occurrence of one given the other has occurred

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Conditional Probability

□ Given an event A and a set of mutually exclusive and exhaustive events B_i

□ Law of Total Probability

$$P(B_1|A) + P(B_2|A) + \cdots P(B_n|A) = 1$$

Independence

□ A and B are independent if knowing that one occurs does not change the probability that the other one occurs.

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

- Equivalently

$$P(A \cap B) = P(A)P(B)$$

Exercise 1

□ The BUAN 301 mid-term is a multiple-choice exam where each question has 4 options and exactly one option is right. The student either knows the answer or chooses between guessing the answer and copying it from the neighbour. She is an average student, so the probability of her knowing the right answer is known to be 0.6. She is morally upright so in the event of her not knowing the answer she chooses to guess with a probability of 0.3. Her last option is to sneak a look into the neighbour's paper and copy the answer from there. In this case because her neighbour is the topper she has a probability of 0.7 of getting it right.

□ Question: What is the probability that she will get an answer right?

Approach

- ☐ Relevant Events?
- ☐ Probabilities of the relevant events?
- ☐ Conditional Probabilities of the relevant events?
- ☐ Identify what probability value we seek?
- ☐ Compute and report.

Bayes' Theorem

□ Given two events A and B and the conditional probabilities as defined

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B|A) = \frac{P(A \cap B)}{P(A)}$$

□ Bayes' Theorem states that

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Bayes' Theorem

□ When we are dealing with mutually exclusive and exhaustive events A_i along with another event B

□ The Conditional Probability

$$P(A_k|B) = \frac{P(A_k)P(B|A_k)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \cdots P(A_n)P(B|A_n)}$$

Exercise 2

Black boxes used in aircrafts manufactured by three companies A , B and C . 75% are manufactured by A , 15% by B , and 10% by C . The defect rates of black boxes manufactured by A , B , and C are 4%, 6%, and 8%, respectively.

❑ Question

- If a black box tested randomly is found to be defective, what is the probability that it is manufactured by company A ?

Solution

□ Let $P(A)$, $P(B)$, $P(C)$ be probabilities corresponding to the black box being manufactured by companies A, B, and C, respectively, and $P(D)$ be the probability of the black box being defective. We are interested in calculating the probability $P(A|D)$.

□ Given

- $P(D|A) = 0.04, P(A) = 0.75$
- $P(D|B) = 0.06, P(B) = 0.15$
- $P(D|C) = 0.08, P(C) = 0.10$

Solution

□ Compute

- $P(D) = P(A)P(D|A) + P(B)P(D|B) + P(C)P(D|C)$
- $P(D) = (0.04 * 0.75) + (0.15 * 0.06) + (0.08 * 0.10) = 0.047$

□ by Bayes Theorem

$$\bullet P(A|D) = \frac{P(D|A)P(A)}{P(D)} = \frac{(0.75)*(0.04)}{0.047} = 0.638$$

References

- ❑ Statistics for Business and Economics : (Anderson/Sweeney/Williams/Camm/Cochran)
 - Chapter 4 : Introduction to Probability

- ❑ Business Analytics : The Science of Data-driven Decision Making (*U. Dinesh Kumar*)
 - Chapter 3: Introduction to Probability
 - till Section 3.5

Next Class

□ Distribution, Random Variables

□ Reading: Statistics for Business & Economics (*Anderson/Sweeney/Williams/ Camm/ Cochran*)

- Chapter 5: Discrete Probability Distributions

□ Reading: First Course in Probability (*Sheldon Ross*)

- Chapter 4: Random Variables

Thank You!

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